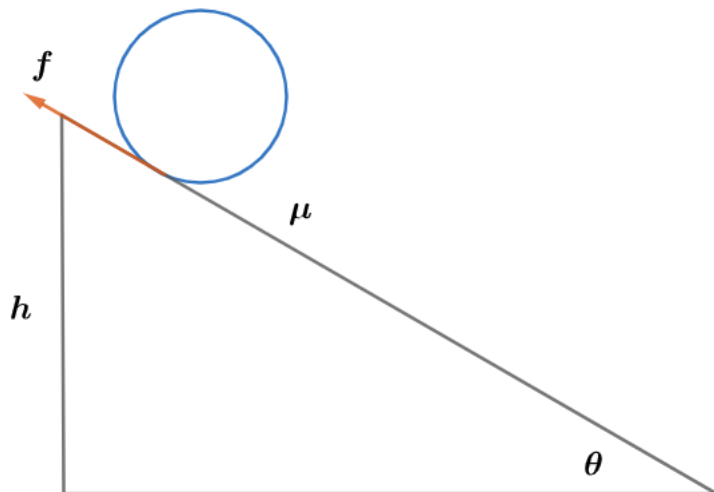


2021 F=ma Exam: Problem 21

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For rolling without slipping, recall the acceleration of a rolling object down an incline is $a = \frac{g \sin \theta}{1 + \beta}$ where the object has moment of inertia $I = \beta mr^2$. Applying $\tau = I\alpha$, we have

$$fr = I\alpha = (\beta mr^2) \frac{a}{r} = \beta mr \left(\frac{g \sin \theta}{1 + \beta} \right)$$

Using our constraint for static friction,

$$f = \frac{\beta}{1 + \beta} (mg \sin \theta) \leq \mu mg \cos \theta$$

$$\tan \theta \leq \mu \left(1 + \frac{1}{\beta} \right)$$

In our case,

$$\tan \theta_c = (0.2) \left(1 + \frac{1}{2/5} \right) = 0.7$$

$$\theta_c = 35^\circ$$

Thus, the answer is A since that is the only angle smaller than θ_c so no energy will be dissipated.